

Phase 3 — Conditional Volatility: ARCH / GARCH Modelling

Complete Analysis of Results

Quantitative Energy Research
Mathematical Finance RAship

Input: `ar4_residuals_phase2.csv` from Phase 2
Sample: 1,462 daily observations, 2015–2018

Based on: Engle (1982) · Bollerslev (1986) · Tol (1997)

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Overview and Purpose of Phase 3

Phase 3 receives as its sole input the AR(4) residuals $\hat{\varepsilon}(t)$ produced by Phase 2. After fitting the AR(4) model, the residuals are serially uncorrelated in their conditional mean but exhibit *conditional heteroskedasticity*: their variance is not constant over time but clusters into prolonged periods of elevated or suppressed volatility. This phenomenon was identified in Phase 2 by the Ljung-Box test on $\hat{\varepsilon}^2(t)$ and the ARCH-LM test of Engle (1982), both of which rejected homoskedasticity at the 0.1% level.

The objective of Phase 3 is to model this residual volatility clustering through the GARCH(1,1) framework of Bollerslev (1986), and to construct the two-component total conditional variance:

$$\sigma_{\text{total}}^2(t) = \sigma_{\text{seasonal}}^2(t) \times h(t), \quad (1)$$

where $\sigma_{\text{seasonal}}^2(t)$ is the deterministic Fourier-fitted seasonal variance from Phase 1 and $h(t)$ is the stochastic GARCH(1,1) conditional variance multiplier estimated in this phase. This decomposition, introduced for wind speed by Tol (1997), separates the predictable seasonal amplitude of wind variability from the unpredictable, time-varying clustering component. Both components are required for the derivatives pricing formulae in Phase 6.

Phase 3 proceeds through five logical steps matching the roadmap:

1. Re-confirm ARCH effects on the Phase 2 residuals via ARCH-LM and Ljung-Box tests on $\hat{\varepsilon}^2(t)$.
2. Estimate GARCH(1,1) by Gaussian MLE; verify $\alpha + \beta < 1$.
3. Construct and analyse the two-component model (1).
4. Validate the standardised GARCH residuals $z(t) = \hat{\varepsilon}(t)/\sqrt{h(t)}$ for approximate i.i.d. $\mathcal{N}(0, 1)$ behaviour.
5. Assess the value added by GARCH via interval forecast coverage and the log proper scoring rule; compute volatility persistence.

Data Loading (Cell 1)

All inputs are loaded from CSV files saved at the end of prior phases:

- `ar4_residuals_phase2.csv` — 1,462 daily AR(4) residuals, 2014-12-31 to 2018-12-31, zero NaN values.
- `ar4_coefficients.csv` — the four AR coefficients $\hat{\beta}_1, \dots, \hat{\beta}_4$ for reference.
- `W_tilde_phase1.csv` — the standardised series $\tilde{W}(t)$, used to reconstruct $\sigma_{\text{seasonal}}^2(t)$.

The seasonal variance $\sigma_{\text{seasonal}}^2(t)$ is reconstructed from the Phase 1 Fourier coefficients:

$$\sigma_{\text{seasonal}}^2(t) = c_0 + c_1 \cos\left(\frac{2\pi t}{365}\right) + c_2 \cos\left(\frac{4\pi t}{365}\right), \quad (2)$$

with $c_0 = 0.13138$, $c_1 = 0.09579$, $c_2 = -0.00332$, giving a range of $[0.0323, 0.2238]$ and $\sigma_{\text{seasonal}}(t) \in [0.180, 0.473]$.

ARCH-LM Entry Diagnostics (Cell 2)

Residual properties before GARCH

Table 1 summarises the AR(4) residual statistics entering Phase 3.

Table 1: Descriptive statistics of AR(4) residuals $\hat{\varepsilon}(t)$ at the Phase 3 entry point.

Statistic	Value	Target
n	1,462	—
Mean	+0.000573	≈ 0
Standard deviation	0.7803	—
Skewness	+0.361	≈ 0
Pearson kurtosis	4.129	3 (Gaussian)
Excess kurtosis	+1.129	≈ 0
JB p -value	0.000	—
KS p -value	0.083	> 0.05

The KS test passes (body of the distribution approximately Gaussian) while the JB test rejects normality, indicating fat tails. This is the canonical fingerprint of ARCH: the conditional distribution is approximately Gaussian but the unconditional distribution has excess kurtosis due to the time-varying variance.

Ljung-Box on $\hat{\varepsilon}(t)$

Table 2: Ljung-Box test on AR(4) residuals $\hat{\varepsilon}(t)$. H_0 : no serial autocorrelation. Pass requires $p > 0.05$.

Lags	LBQ statistic	p -value	Verdict
5	1.199	0.945	Pass
10	2.896	0.984	Pass
15	9.396	0.856	Pass
20	19.242	0.506	Pass

The AR(4) mean equation remains correctly specified: all serial autocorrelation is removed at every lag length tested.

Ljung-Box on $\hat{\varepsilon}^2(t)$ and ARCH-LM test

Table 3: Ljung-Box test on squared residuals $\hat{\varepsilon}^2(t)$. H_0 : homoskedastic. $p < 0.05$ indicates ARCH effects.

Lags	LBQ statistic	p -value	Verdict
5	22.975	0.0003	ARCH effects
10	31.819	0.0004	ARCH effects
15	36.705	0.0014	ARCH effects
20	41.382	0.0033	ARCH effects

The Engle (1982) ARCH-LM test confirms this independently:

$$\begin{aligned} \text{ARCH-LM}(5) &= 18.80, & p &= 0.0021, & R^2 &= 0.013, \\ \text{ARCH-LM}(10) &= 23.23, & p &= 0.0099, & R^2 &= 0.016. \end{aligned}$$

ARCH effects are confirmed at all lag lengths and by both test procedures. GARCH(1,1) estimation is the appropriate next step.

GARCH(1,1) Estimation (Cell 3)

Model specification and estimation

The GARCH(1,1) model of Bollerslev (1986) specifies:

$$\hat{\varepsilon}(t) = \sqrt{h(t)} \cdot z(t), \quad z(t) \sim \mathcal{N}(0, 1) \text{ i.i.d.}, \quad (3)$$

$$h(t) = \omega + \alpha \hat{\varepsilon}^2(t-1) + \beta h(t-1). \quad (4)$$

with non-negativity constraints $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$ and covariance stationarity condition $\alpha + \beta < 1$ (Bollerslev 1986, Theorem 1). Parameters are estimated by Gaussian quasi-maximum likelihood:

$$\ell(\omega, \alpha, \beta) = -\frac{1}{2} \sum_{t=1}^n \left[\ln h(t) + \frac{\hat{\varepsilon}^2(t)}{h(t)} \right], \quad (5)$$

with $h(1)$ initialised at the sample unconditional variance (Tol 1997 initialisation). Optimisation uses the SLSQP constrained algorithm with inequality constraints enforcing all three parameter restrictions.

Estimated parameters

Table 4: GARCH(1,1) maximum likelihood estimates. Significance: *** $p < 0.01$, * $p < 0.10$.

Param.	Estimate	Std. Err.	t -stat	p -value	Sig.
$\hat{\omega}$	0.010044	0.005347	+1.878	0.060	*
$\hat{\alpha}$	0.031050	0.008373	+3.709	0.000	***
$\hat{\beta}$	0.952421	0.013926	+68.39	0.000	***
Log-likelihood = -348.14; AIC = 702.28; BIC = 718.14					
$\hat{\alpha} + \hat{\beta} = 0.9835 < 1 \Rightarrow$ covariance stationary ✓					

Interpretation

Stationarity. The persistence $\hat{\alpha} + \hat{\beta} = 0.9835$ is strictly less than one, confirming that $h(t)$ is a covariance-stationary process with a well-defined unconditional variance:

$$\frac{\hat{\omega}}{1 - \hat{\alpha} - \hat{\beta}} = \frac{0.01004}{0.0165} = 0.6077 \approx \mathbb{E}[\varepsilon^2] = 0.6088 \quad \checkmark \quad (6)$$

The near-exact agreement between the GARCH implied long-run variance and the sample variance confirms that the MLE converged to the correct global optimum.

$\hat{\beta} \gg \hat{\alpha}$. The ratio $\hat{\beta}/\hat{\alpha} = 30.7$ indicates that wind volatility is overwhelmingly driven by the carryover of past variance rather than the impact of fresh shocks. A shock today raises tomorrow's conditional variance by only 3.1% (the $\hat{\alpha}$ contribution), but 95.2% of today's variance level persists to tomorrow regardless ($\hat{\beta}$). This structure is described by Tol (1997) as typical of daily meteorological series, where volatility is governed by slow-moving synoptic weather regimes rather than the sharp, episodic jumps observed in financial returns.

Volatility persistence and half-life. The half-life of a volatility shock is the time required for $h(t)$ to decay halfway back to its long-run level after a one-unit impulse. Following Bollerslev (1986, Section 3):

$$\tau_{1/2} = \frac{-\ln 2}{\ln(\hat{\alpha} + \hat{\beta})} = \frac{-\ln 2}{\ln(0.9835)} \approx 41.6 \text{ days}. \quad (7)$$

This is substantially longer than the mean-reversion half-life of wind speed anomalies ($\tau_{1/2} = 2.0$ days, Phase 2). The distinction is physically meaningful: the *level* of wind speed reverts to its seasonal mean within days, but the *regime of volatility* — whether the atmosphere is in a high-variability or low-variability period — persists for weeks. This is consistent with the multi-week persistence of large-scale atmospheric circulation patterns such as blocking events and the North Atlantic Oscillation (NAO) phase.

Table 5 shows the impulse-response of $h(t)$ to a unit shock.

Table 5: Impulse-response of GARCH(1,1) conditional variance $h(t)$ to a unit shock at $t = 0$. Values show $(\hat{\alpha} + \hat{\beta})^k$.

Lag (days)	$h(t)/h(0)$	% remaining
0	1.0000	100.0%
5	0.9200	92.0%
10	0.8465	84.6%
20	0.7165	71.6%
42	≈ 0.5	50.0% (half-life)
90	0.2234	22.3%

Two-Component Volatility Model (Cell 4)

Decomposition structure

The total conditional variance of the Box-Cox transformed wind speed deviations $[W^\lambda(t) - \Lambda(t)]$ factorises as:

$$\sigma_{\text{total}}^2(t) = \underbrace{\sigma_{\text{seasonal}}^2(t)}_{\text{Phase 1: deterministic}} \times \underbrace{h(t)}_{\text{Phase 3: stochastic}}. \quad (8)$$

This factorisation follows directly from the Phase 1 standardisation $\tilde{W}(t) = [W^\lambda(t) - \Lambda(t)]/\sigma(t)$, the AR(4) mean model $\tilde{W}(t) = \sum_k \hat{\beta}_k \tilde{W}(t-k) + \hat{\varepsilon}(t)$, and the GARCH model $\hat{\varepsilon}(t) = \sqrt{h(t)} \cdot z(t)$, which together imply $\text{Var}_t[W^\lambda(t) - \Lambda(t)] = \sigma^2(t) \cdot h(t)$.

Numerical results

Table 6: Monthly averages of the two-component volatility decomposition. σ_{seas}^2 is the Phase 1 Fourier seasonal variance; $\bar{h}$ is the monthly mean of the GARCH conditional variance; σ_{total}^2 and σ_{total} are their product and square root.

Month	σ_{seas}^2	$\bar{h}(t)$	σ_{total}^2	σ_{total}
Jan	0.2198	0.5470	0.1202	0.3462
Feb	0.1984	0.6032	0.1197	0.3439
Mar	0.1598	0.6680	0.1070	0.3260
Apr	0.1108	0.6780	0.0754	0.2729
May	0.0652	0.7580	0.0496	0.2209
Jun	0.0374	0.7680	0.0287	0.1677
Jul	0.0370	0.7252	0.0266	0.1618
Aug	0.0646	0.5727	0.0367	0.1903
Sep	0.1100	0.5039	0.0555	0.2332
Oct	0.1591	0.4652	0.0739	0.2705
Nov	0.1988	0.5227	0.1041	0.3213
Dec	0.2203	0.5220	0.1150	0.3380

Physical interpretation

The two components exhibit an interesting seasonal interplay. The Phase 1 seasonal variance $\sigma_{\text{seasonal}}^2(t)$ is highest in winter (Jan–Feb: ≈ 0.22) and lowest in summer (Jun–Jul: ≈ 0.037), reflecting the larger amplitude of synoptic wind events in the cold season. The GARCH component $\bar{h}(t)$ shows the opposite pattern: it is elevated in May–June (≈ 0.76) and lower in October (≈ 0.47).

These two effects partially cancel in the total variance $\sigma_{\text{total}}^2(t)$, which is highest in January–February (≈ 0.12) and lowest in July (≈ 0.027). The winter peak in total volatility is driven by the dominant seasonal component, while the GARCH multiplier modulates the magnitude. This compensation structure implies that pricing wind derivatives purely from the seasonal variance (ignoring $h(t)$) would overestimate summer volatility and underestimate the GARCH contribution in transitional months.

Pricing approximation

Following Tol (1997), for derivatives pricing over a horizon $[\tau_1, \tau_2]$, the GARCH multiplier $h(t)$ is treated as piecewise-constant at its current estimated value $h(t_0)$:

$$h(t) \approx h(t_0) = 0.4137 \quad \text{for all } t \in [\tau_1, \tau_2]. \quad (9)$$

This avoids the need to forecast $h(t)$ over the full pricing horizon and provides a conservative (slightly wide) pricing interval. Phase 4 will use this as the initial condition for the CAR(4) volatility term structure.

Standardised Residual Validation (Cell 5)

Theoretical requirements

Under correct GARCH(1,1) specification, the standardised residuals $z(t) = \hat{\varepsilon}(t)/\sqrt{h(t)}$ must be approximately i.i.d. $\mathcal{N}(0, 1)$. Specifically: (a) no serial autocorrelation in $z(t)$ — Ljung-Box on $z(t)$, $p > 0.05$; (b) no remaining ARCH effects in $z^2(t)$ — Ljung-Box on $z^2(t)$, $p > 0.05$; (c) approximately normal — skewness ≈ 0 , Pearson kurtosis ≈ 3 . Bollerslev (1986) notes that a kurtosis slightly above 3 is acceptable in finite samples even under correct specification.

Descriptive statistics before and after GARCH

Table 7: Comparison of residual distribution before and after GARCH standardisation. Target values apply to $z(t)$.

Statistic	$\hat{\varepsilon}(t)$ before	$z(t)$ after	Target
Mean	+0.00057	+0.00318	≈ 0
Std deviation	0.7803	1.0006	≈ 1
Skewness	+0.3608	+0.4512	≈ 0
Kurtosis (Pearson)	4.1292	4.1162	3
Excess kurtosis	+1.1292	+1.1162	≈ 0
JB p -value	0.000	0.000	> 0.05
KS p -value	0.083	0.079	> 0.05

Ljung-Box on $z(t)$ and $z^2(t)$

Table 8: Ljung-Box tests on $z(t)$ (serial autocorrelation) and $z^2(t)$ (remaining ARCH effects). Both pass at all lag lengths.

Lags	$z(t)$ — serial corr.		$z^2(t)$ — ARCH effects	
	LBQ	p -value	LBQ	p -value
5	1.344	0.930	2.210	0.819
10	2.594	0.989	3.103	0.979
15	8.200	0.916	6.006	0.980
20	15.202	0.765	8.477	0.988

The ARCH-LM test on $z(t)$ gives $\text{ARCH-LM}(5) = 2.198$, $p = 0.821$ and $\text{ARCH-LM}(10) = 3.088$, $p = 0.979$. All residual ARCH effects are removed by the GARCH(1,1) model.

Tail behaviour and remaining non-normality

The standardised residuals have $|z| > 2\sigma$: 69 observations (4.7%, expected 4.6% ✓), but $|z| > 3\sigma$: 12 observations (0.82%, expected 0.3%). The right tail is heavier than Gaussian. Table 9 shows the five largest events.

Table 9: Top-5 standardised GARCH residuals $z(t)$ by absolute value.

Date	$z(t)$	Month
2017-06-28	+5.785	June
2017-12-10	+3.863	December
2016-07-19	-3.623	July
2016-03-31	+3.534	March
2018-10-29	+3.344	October

The June 2017 event ($z = +5.785$, or 5.8 conditional standard deviations) represents an extreme wind anomaly that the Gaussian GARCH model cannot fully absorb. The persistence of kurtosis = 4.12 and JB rejection after GARCH standardisation indicates that the conditional error distribution has fatter tails than the Gaussian. This is a documented finding for meteorological GARCH models (Tol 1997) and motivates the optional replacement of Gaussian noise with a Student- t or NIG Lévy process in Phase 4. For the purposes of pricing wind derivatives with Gaussian dynamics, the GARCH specification is adequate: the ARCH effects are fully removed, and the extreme events are infrequent.

Interval Forecast Comparison and Persistence (Cell 6)

Interval forecast design

Since GARCH(1,1) does not alter the conditional mean of $\hat{\varepsilon}(t)$, the point-forecast RMSE is identical for AR(4) and AR(4)+GARCH. The appropriate performance measure is the *interval forecast*, which evaluates whether the model correctly characterises its own uncertainty. A 95% prediction interval from a correctly specified model should contain approximately 95% of realised observations.

The test set comprises 220 observations (the final 15% of the sample). Two models are compared:

- **AR(4) homoskedastic:** fixed 95% interval $\pm 1.96\hat{\sigma}_{\text{train}}$ using the training-set residual standard deviation.
- **AR(4)+GARCH(1,1):** adaptive 95% interval $\pm 1.96\sqrt{h(t)}$ using the time-varying conditional standard deviation.

Results

Table 10: Interval forecast performance. Test set: 220 observations. Target coverage: 95%. Log-score: higher (less negative) is better.

Model	Coverage	Mean width	Log-score	Calibration
AR(4) homoskedastic	97.3%	3.167	−0.982	Overcovering
AR(4)+GARCH(1,1)	96.4%	2.654	−0.915	Pass

The log-score gain from GARCH is +0.067 per observation. The GARCH model achieves better calibration in two ways. First, it narrows the average interval width from 3.167 to 2.654 (a 16% reduction in mean width), producing more informative intervals during calm periods. Second, it correctly widens the interval during volatile periods, preventing the undercoverage that would result from using fixed- σ bands during clustered high-variance episodes.

The AR(4) homoskedastic model overcovering at 97.3% indicates that its fixed bands are on average too wide — they are calibrated to the full-sample variance, which includes high-volatility episodes. The GARCH model adapts the interval width to the current volatility regime and achieves coverage closer to the nominal 95% target.

Volatility persistence summary

Volatility persistence. $\hat{\alpha} + \hat{\beta} = 0.9835$. Impulse decay: a volatility shock retains 92.0% of its magnitude after 5 days, 84.6% after 10 days, and 50% after 42 days. Long-run unconditional variance = 0.6077, matching the empirical $\mathbb{E}[\varepsilon^2] = 0.6088$. The ratio $\hat{\beta}/\hat{\alpha} = 30.7$ confirms that wind volatility is persistent but not spiky: it is governed by slow atmospheric circulation regimes (NAO, blocking) rather than sharp episodic shocks.

Diagnostic Plots

Figure 1 presents the six-panel Phase 3 diagnostic figure.



Figure 1: Phase 3 — GARCH(1,1) conditional volatility decomposition. *Row 1:* AR(4) residuals with GARCH adaptive 95% confidence bands (orange) and fixed- σ bands (red dashed). The GARCH bands visibly widen around the 2016–2017 high-volatility episode and the June 2017 extreme event. *Row 2:* GARCH conditional variance $h(t)$ (crimson) with unconditional variance (navy dashed). The high-persistence structure ($\alpha + \beta = 0.983$) produces slowly-evolving volatility regimes. *Row 3:* Two-component total volatility $\sigma_{\text{total}}(t)$ (green) vs Phase 1 seasonal baseline $\sigma_{\text{seasonal}}(t)$ (orange dashed). The seasonal component dominates the annual pattern; the GARCH multiplier modulates the amplitude within each year. *Row 4, left:* Histogram of standardised residuals $z(t)$ vs $\mathcal{N}(0,1)$ — body well-fitted, right tail heavy. *Row 4, centre:* Q-Q plot of $z(t)$ — linear in the body, right tail deviation at the June 2017 extreme. *Row 4, right:* Monthly mean $\sigma_{\text{seasonal}}^2$ and $h(t)$ — the opposing seasonal patterns partially cancel in the total variance.

Phase 3 Complete Summary

Complete Phase 3 results.

[1] **ARCH-LM entry diagnostic.** ARCH-LM(5) = 18.80, $p = 0.002$. LBQ($\hat{\varepsilon}^2$, 20): $p = 0.003$. ARCH effects confirmed at all lag lengths $\rightarrow$ GARCH(1,1) estimation justified.

[2] **GARCH(1,1) estimated parameters.** $\hat{\omega} = 0.01004$ (*), $\hat{\alpha} = 0.03105$ (***), $\hat{\beta} = 0.95242$ (***). $\hat{\alpha} + \hat{\beta} = 0.9835 < 1 \rightarrow$ stationary $\checkmark$. Volatility half-life = 41.6 days. AIC = 702.28; BIC = 718.14.

[3] **Two-component model** $\sigma_{\text{total}}^2(t) = \sigma_{\text{seasonal}}^2(t) \times h(t)$. $\sigma_{\text{seasonal}}^2$ range: [0.032, 0.224]. $h(t)$ range: [0.314, 1.285]. σ_{total}^2 range: [0.016, 0.194]. Current $h(t_0) = 0.4137 \rightarrow$ Phase 4 pricing input.

[4] **Standardised GARCH residual validation** $z(t) = \hat{\varepsilon}/\sqrt{h}$. LBQ(z^2 , 20): $p = 0.988 \rightarrow$ ARCH fully removed $\checkmark$. ARCH-LM(5) on z : $p = 0.821 \rightarrow$ no residual clustering $\checkmark$. Skewness = +0.451; kurtosis = 4.116; JB $p = 0.000$; KS $p = 0.079$. Residual non-normality (fat right tail) driven by June 2017 extreme ($z = +5.785$); Gaussian GARCH adequate for pricing, Student- t or NIG extension noted as future refinement.

[5] **Interval forecast (test set: 220 obs).** AR(4)+GARCH coverage = 96.4% (target 95%, calibrated $\checkmark$) vs AR(4) fixed- $\sigma = 97.3\%$ (overcovering). Log-score gain from GARCH: +0.067 per observation. Mean interval width reduction: 16%. Persistence $\hat{\alpha} + \hat{\beta} = 0.9835$.

Phase 3 outputs: `garch_h_series_phase3.csv` ($h(t)$, 1,462 obs), `garch_z_series_phase3.csv` ($z(t)$, 1,462 obs), `garch_parameters_phase3.csv` ($\hat{\omega}, \hat{\alpha}, \hat{\beta}$), `sigma2_total_phase3.csv` ($\sigma_{\text{total}}^2(t)$, 1,462 obs).

Ready for Phase 4 — CAR(4) continuous-time embedding. Key inputs: `ar4_coefficients.csv` ($\beta_1 \dots \beta_4 \rightarrow \alpha_1 \dots \alpha_4$ mapping), `garch_h_series_phase3.csv` ($h(t_0) = 0.4137$ as piecewise-constant pricing volatility).